PALAEOMAGNETISM OF DIABASE DYKES OF THE CANADIAN SHIELD[1]

W. F. Fahrig, E. H. Gaucher, and A. Larochelle
Geological Survey of Canada, 601 Booth Street, Ottawa, Canada
Received November 10, 1964

ABSTRACT

Some areas of the Canadian Shield are profusely intruded by swarms of subparallel basaltic dykes. These typically have a diabasic texture and formerly were considered to be strictly the intrusive equivalents of tholeiitic flood basalt.

About 650 oriented samples were collected from about 25 dyke swarms and preliminary palaeomagnetic data from six of them are presented here. The Mackenzie swarm has a K–Ar age of 1 295 million years, occurs throughout the western Canadian Shield, and is the most extensive swarm of basic dykes known anywhere in the world. The other swarms are the Molson dykes (1 445 m.y.) in northeastern Manitoba, the Marathon dykes (1 810 m.y.) just north of Lake Superior, the Sudbury dykes (1 285 m.y.) of southeastern Ontario, the Matachewan dykes (2 485 m.y.) of western Quebec and northeastern Ontario, and the Abitibi dykes (1 230 m.y.), which occur from Sault Ste. Marie, Ontario, to Lake Mistassini, Quebec.

The palaeomagnetic virtual pole positions of six swarms were derived from the mean of their measured remanent magnetization directions. These directions of magnetization were determined from the samples after they were magnetically washed in an a-c. field of 80 or 130 oersteds. The reasons for the dispersion within the individual swarms are discussed. The pole positions of the Sudbury and Mackenzie dykes are almost identical and the two swarms are the same age within the limits of analytical uncertainty. However, basalt of the Sudbury swarm is alkalic and more undersaturated than basalt of the Mackenzie dykes.

No evidence was found to contradict the usual assumption that unaltered basic igneous rocks of this type acquire and generally retain a stable magnetization which was parallel with the earth's magnetic field at the time of the dyke intrusion.

INTRODUCTION

This is the initial report on the results of an extensive study of the magnetic, and in particular of the palaeomagnetic properties of basaltic dykes of the Canadian Shield. These intrusives typically exhibit a diabasic texture and for this reason are commonly referred to simply as diabase dykes. Such rocks appear to be ideal subjects for palaeomagnetic study because some swarms extend over large areas of the Shield; the intrusion of an individual swarm is thought to have been an igneous event of short duration; they can be dated by the K–Ar method; they have intruded the Shield at numerous stages throughout the span of Precambrian time; and many lack any indication of having been metamorphosed or disturbed since they were emplaced.

The first attempt to investigate and interpret the remanent magnetism of diabase dyke swarms of the Canadian Shield was made by D. W. Strangway (1961). He studied swarms in eastern Ontario and southwestern Quebec and concluded that the most stable component of magnetization in diabase dykes lies close to the plane of the dyke. He concluded from these results and a review of literature that a diabase dyke can be treated as an infinite cooling sheet

[1]Published with permission of the Director, Geological Survey of Canada.

which, by virtue of its geometry, sets up a demagnetizing field within the dyke that tends to cancel the component of magnetization normal to the dyke.

There are more than 20 major, readily recognizable diabase dyke swarms in the Canadian Shield. The present preliminary report deals with the palaeomagnetism of 6. The study of the remaining swarms is in progress and will be reported on at a later date.

The usual assumptions made in the investigation of the palaeomagnetism of igneous rocks is that the rocks under investigation contain a stable component of magnetization that is parallel with the earth's magnetic field at the time of their formation and that the earth's field configuration may be considered to be that of a coaxial geocentric dipole over the period of time involved in the cooling of the dyke swarm. Our immediate objectives are then threefold: to examine this assumption, to derive a pole position from the thermoremanent magnetization of each swarm and study variations of palaeomagnetic directions and intensities within apparently homogeneous swarms, and from this to assess palaeomagnetism as a tool in the correlation of basic dykes.

SAMPLE COLLECTION AND MEASUREMENT

Six hundred and fifty oriented samples were collected from about 350 sites in the initial phase of the program; 225 of these samples are from the six swarms discussed in this report. At each site an attempt was made to collect at least two samples. It would have been desirable to collect more oriented samples at each site, but the additional time required would have meant that the swarms could not have been sampled throughout, a requirement for a concurrent systematic geochemical and petrographic study of the swarms.

Two 1-in. cubes were cut from each sample and their magnetization was determined with either a spinner or astatic magnetometer. To eliminate the viscous component of magnetization usually present in igneous rocks, all cubes were subjected to an alternating field whose peak value was empirically set at 80 or 130 oersteds. Since the apparatus (Larochelle 1962) used in this operation did not allow tumbling of the specimens during treatment, the demagnetizing procedure was repeated three times with the three axes of the cube successively aligned parallel with the axis of the demagnetizing coil. The two cubes of any individual sample were purposely oriented in opposite positions (viz. $+x$, $-y$, $-z$ vs. $-x$, $+y$, $+z$) to detect the introduction of an anhysteric component in the sample. Since in 14% of the samples the magnetic vectors of the two cubes cut from the individual samples diverged by more than 30 degrees after the treatment, these samples were temporarily considered 'unstable'. In most samples, this apparent 'instability' was the result of the introduction of an anhysteric component of magnetization. Some of these 'unstable' samples have since been rewashed with improved equipment; the anhysteric component introduced into the samples was eliminated and the magnetization directions obtained conform well with results obtained on the original 'stable' samples. For the purpose of this preliminary paper, however, the magnetic vectors of

only the 'stable' samples have been used in calculating the mean magnetization directions. The magnetization direction for each 'stable' sample was obtained by averaging the measurements made on the two cubes cut from that sample.

Generally, the dispersion of the magnetic vectors within a swarm decreased markedly after the alternating field treatment.

STATISTICAL TREATMENT OF DATA

The average palaeomagnetic direction of stable samples of each swarm, the precision parameter k, and the radius α of the circle of confidence about the mean at the probability level of 95% were calculated by the method introduced by Fisher (1953). For the Marathon swarm for which only 11 samples were available, two 'anomalous' directions were excluded in the computation of the mean because of their pronounced distorting effect.

The stable samples of each swarm appear to represent two populations. One exhibits a normal Fisherian distribution with a precision parameter of about 20 and the other comprises randomly oriented samples. The latter population forms 10 to 25% of the samples in a swarm and the samples making up this population will be referred to hereafter as 'anomalous'. The magnetization directions of the Abitibi swarm may be used in examining the above statement. A histogram representing the deviations ψ of the 58 individual observations from the mean is represented in Fig. 1*a*. The k calculated for this population was 3.5 and the histogram of ψ assuming a Fisherian distribution for the 58 samples is represented in Fig. 1*b*. The fit of the observed data to the theoretical was tested by a χ^2 test (Watson and Irving 1956) and it was found that the distribution of the samples was not Fisherian at the 0.01% probability level (Table I). On the other hand if it is assumed that 75% of the samples (44) belong to a normal population having a k of 20 and the remaining 14 samples are random ($k = 0$) the histogram of Fig. 1*c* is obtained. The χ^2 test shows that this distribution fits the observed data much better (Table II).

The magnetization measurements of the other swarms are characterized by similar behavior. The values of k listed in Table I would thus be conservative estimates, and the same remark applies to the values of α for the radius of confidence, since these populations are all contaminated with random samples. It was also noted that if two samples from a site, collected more than 20 ft apart, have magnetization directions within 30 degrees of one another, almost invariably they are within 30 degrees of the mean magnetization direction of the swarm to which they belong. If two such samples differ in magnetization direction by more than 30 degrees, then 4 times out of 5, the magnetization direction of one sample will lie within 30 degrees of the mean direction of the swarm whereas the direction of the second sample is completely erratic, i.e. is considerably more than 30 degrees from the mean. This fact also suggests two populations, as postulated earlier.

The measurements may suggest the existence of a primary magnetic population of normal distribution upon which lightning has imposed a second population of random orientation. About 25% of Abitibi samples are randomly

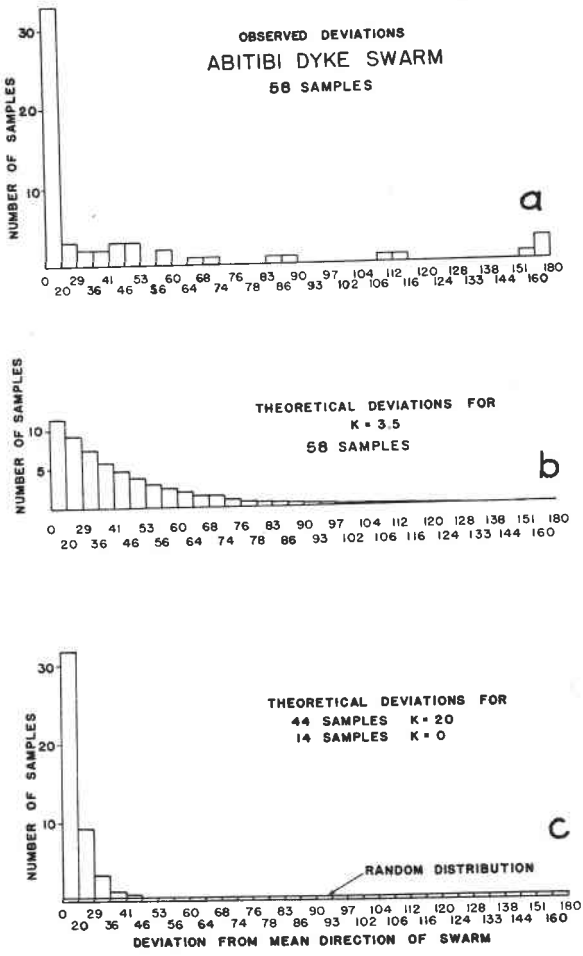


FIG. 1, *a-c*. Observed and theoretical angular deviations from the mean of magnetization directions of the Abitibi dyke samples.

oriented while only 10% of Mackenzie samples are random. The Mackenzie dykes lie in a different climatic belt and could have experienced fewer strokes of lightning. Further demagnetization treatment under alternating fields of higher peak amplitudes will be carried out in an attempt to clarify this point.

There are a number of ways in which the mean magnetization directions might be more accurately calculated by statistically eliminating the anomalous measurements. For example, the mean magnetization direction of the Abitibi dykes was recalculated using only those sites in which the magnetization direction of the two samples differ by less than 30 degrees (Table I). The mean thus obtained differs by only 2 degrees from the mean obtained by averaging all of the stable samples from this swarm. However, the precision parameter increases from 3.5 to 35 as a result of this manipulation.

TABLE I

Summary of palaeomagnetic data of Precambrian diabase dykes of the Canadian Shield

Dyke swarm	Mean declination	Mean inclination	N	R	k	α	Pole position		Ellipse of confidence	
							Long.	Lat.	$d\chi$	$d\psi$
Abitibi	264.6	66.6	58	41.5	3.5	11.9	128.8 W.	31.5	19.6	16.1
Abitibi (av. of those sites where magnetization directions of two samples agree within 30°)	268.5	65.1	14*	13.6	35.0	6.8	131.8 W.	32.3	11.0	8.9
Mackenzie	250.8	24.7	34	29.1	6.8	10.2	177.3 W.	3.8	11.0	5.9
Sudbury	264.1	.7	37	28.4	4.1	13.2	166.5 W.	−3.77	13.18	6.59
Molson	209.6	79.9	5	4.8	27.5	14.8	108.9 W.	36.2	28.4	27.2
Marathon (2 anomalous samples excluded from av.)	268.2	60.5	9	8.7	24.8	10.5	146.8 W.	29.0	16	12.2
Matachewan (southwest magnetization)	211.9	−5.9	39	30.3	4.4	12.4	120.8 W.	−37.2	12.5	6.3
Matachewan (northeast magnetization)	16.6	42.3	15	12.6	5.7	17.5	61.2 E.	62.9	21.6	13.3

*Number of sites.
N Number of samples averaged (includes all stable samples except where noted).
R Vector sum of N directions; k, precision parameter; α, semiangle of cone of confidence; $d\chi$, $d\psi$, semiaxes of oval of confidence around each pole position.

TABLE II

χ^2 test for Fisher distribution of deviation ψ (Abitibi samples)

	Range of ψ in degrees					
Frequency	0–20.4	20.4–29.0	29.0–60	60–180	χ^2	Significance
Observed	33	3	12	10	53.6	Yes
Expected*	11.4	9.2	27.3	10.1		0.01% probability level
Observed	33	3	12	10	8.6	No
Expected†	31.8	9.2	6.6	10.4		>10% probability level?

*Assuming a normal population of 58 samples with $k = 3.5$, as calculated from the observed population.
†Assuming a normal population of 44 samples, with $k = 20$ and a random population of 14 samples.

Our data suggest that for purposes of correlation at least three samples more than 20 ft apart should be collected from each dyke. If the magnetization directions of these samples agree within 30 degrees, then their mean will virtually always be close to the mean of the swarm to which the dyke belongs. Much of the value of palaeomagnetic measurements in terms of correlation of Precambrian dykes will lie in the value of negative correlation. In other words if the magnetization direction of a dyke differs considerably from that of another body of rock then the two are unlikely to be the same age. On the other hand, the difficulty inherent in establishing polar wandering curves for the Precambrian means that concordant magnetization directions at present provide only supporting evidence of concordant age.

Some of the above statistical generalization regarding the magnetization of Precambrian basic dykes may be appreciably modified with the application of improved magnetic washing techniques.

ABITIBI DYKES

Geological Setting

The Abitibi dykes form a northeasterly trending swarm that extends roughly 500 miles from at least the Sault Ste. Marie (Ontario) area on the southwest to the Lake Mistassini (Quebec) area on the northeast. The name is derived from Abitibi Lake near which the dykes are particularly well exposed.

This swarm contains a relatively small number of dykes, but individual dykes are remarkably large, up to 800 ft wide, and up to several hundred miles long. Most of the larger dykes are shown on the accompanying map (Fig. 2).

The central parts of the wider dykes are very coarse-grained, and locally contain patches of granophyre and up to a considerable percentage of biotite. The chemical analysis reported in Table III is the average of 10 analyses of samples taken from the chilled margins of 10 Abitibi dykes.

Fahrig and Wanless (1964) reported a K–Ar age of 1 230 m.y. on biotite extracted from a coarse phase of Abitibi dyke rock. This is an average of two determinations.

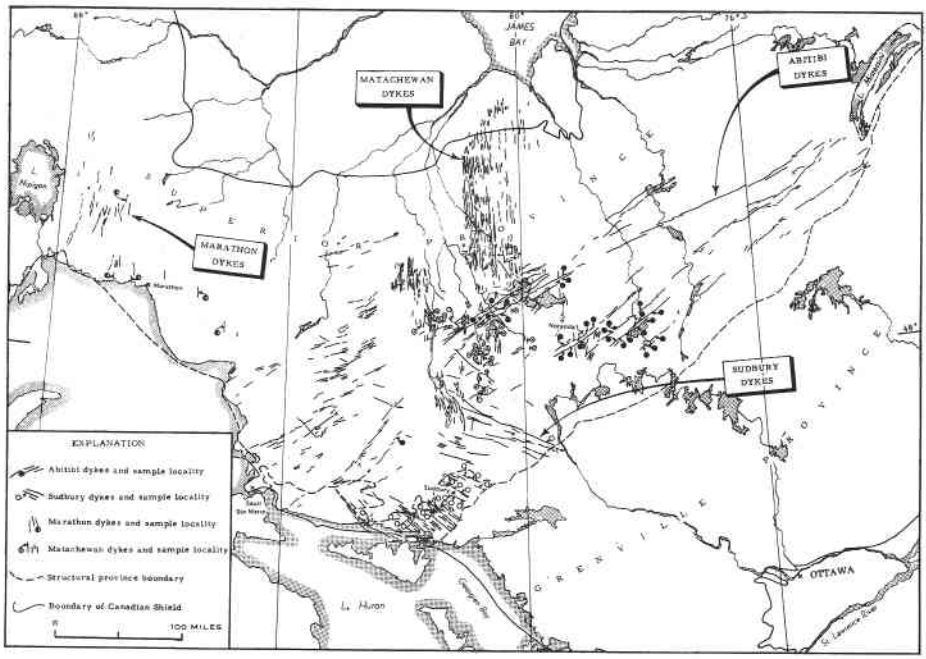


FIG. 2. Map showing the distribution of the Abitibi, Sudbury, Marathon, and Matachewan dykes and location of samples used in this study.

The Abitibi dykes generally appear as strong linear anomalies on aeromagnetic maps of the region between Sault Ste. Marie and Lake Mistassini.

Palaeomagnetic Data

Strangway (1961) published the first palaeomagnetic measurements on Abitibi dykes. These are his 'ENE dykes parallel to Grenville Front'. He reported the averages of the remanent magnetization measurements made on samples from three different sampling localities. His more detailed report (1960) lists 43 sample measurements from four Abitibi dyke localities. His average is within 12 degrees of that reported in this study.

In the present study, 61 samples from 41 sampling sites were studied. The locations of the sites are shown on Fig. 2, and the magnetization directions on Fig. 3. Since, for 6 of the 61 samples, the magnetization vectors of the two cubes cut from a single sample diverged after magnetic washing at 130 oersteds by more than 30°, they were temporarily classified as 'unstable'.

MACKENZIE DYKES

Geological Setting

The Mackenzie dykes form a north-northwesterly striking swarm of basic dykes that occurs virtually throughout the entire western half of the Canadian Shield. They are most abundant in the Slave structural province,[2] which is

[2]For the definition of the structural provinces of the Canadian Shield the reader is referred to Stockwell (1962).

TABLE III

The chemical composition of Canadian Shield diabase dykes and of comparable rocks from the literature

Constituents	Mackenzie 1*	Sudbury 2	Matachewan 3	Marathon 4	Abitibi 5	Av. tholeiite 6	Continental alkalic basalt 7	Continental alkalic basalt 8
SiO_2	47.7	45.9	50.5	50.5	49.6	51.9	50.6	47.1
Al_2O_3	11.9	15.1	13.9	12.7	13.9	15.1	15.4	15.8
Fe_2O_3	3.4	2.9	4.0	3.5	2.4	1.3	4.4	4.0
FeO	12.7	12.1	10.8	10.3	9.9	9.0	6.9	8.3
CaO	9.8	8.5	9.6	9.5	10.3	10.0	8.7	9.4
MgO	4.7	4.2	4.8	4.7	6.8	6.6	7.5	7.4
Na_2O	2.8	3.5	3.0	2.8	2.6	2.1	3.0	3.1
K_2O	0.7	1.2	1.0	1.0	0.8	0.9	1.4	1.3
TiO_2	2.5	3.1	1.4	1.6	1.4	1.3	1.5	2.9
MnO	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.2
CO_2	0.1	0.1	0.1	0.1	0.1			
H_2O^+	1.2	1.7	1.4	2.0		0.9		
H_2O^-						0.5		
P_2O_5						0.2	0.4	0.5
Total	97.7	98.8	100.7	99.0	98.0	100.0	100.0	100.0

*Column No. 1. Average of 20 analyses of samples from margins of 20 Mackenzie Dykes.
2. Average of 10 analyses of samples from margins of 10 Sudbury Dykes.
3. Average of eight analyses of samples from margins of eight Matachewan Dykes.
4. Average of four analyses of samples from margins of four Marathon Dykes.
5. Average of 10 analyses of samples from margins of 10 Abitibi Dykes.
6. Average composition of various tholeiite quartzite diabase occurrences as given by Turner and Verhoogen (1960, p. 215, Table 16, column 9).
7. Newer volcanic series, Victoria, Australia; 27 olivine basalts (A. B. Edwards. The tertiary volcanic rocks of Central Victoria. Geol. Soc. London Quart. J., Vol. 94, p. 309, Nos. 11, 12, 1938).
8. Carboniferous of Scotland; average olivine basalt (27 analyses). (S. I. Tomkeieff. Petrochemistry of the Scottish Carboniferous–Permian igneous rocks. Bull. volcanologique, ser. 2, tome 1, Table 1, No. 13, 1957).

NOTE: Analyses represented by columns 1–5 were carried out in laboratories of the Geological Survey of Canada. Determinations were made as follows:
Si, Al, total Fe, Ca, Mg, K, Ti, Mn by X-ray fluorescence spectroscopy.
Na by flame photometric procedure.
Fe^{2+} titrimetrically after decomposition of the sample in the presence of ammonium vanadate.
CO_2 by an acid-evolution gravimetric method.
H_2O (total) by a modified Penfield method.

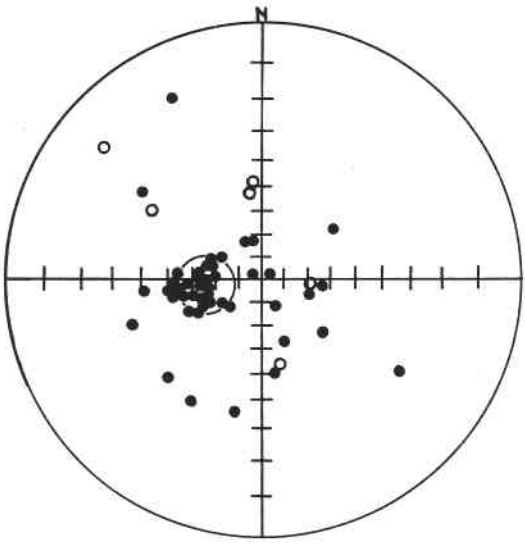


FIG. 3. Stereographic projection of magnetization directions, mean direction, and circle of confidence of Abitibi dyke samples. ● North poles pointing downward; ○, north poles pointing upward.

situated within the Mackenzie District. The dykes of the swarm are called the Mackenzie dykes after the district of that name.

The probable extent of these intrusions has been determined only recently and they are now thought to constitute the most extensive swarm of basic dykes known anywhere in the world, a point illustrated by the accompanying map (Fig. 4). Only the larger of the mapped dykes are shown there and undoubtedly many large dykes remain undiscovered; however, the map probably portrays roughly the regional variation in the density of dyke intrusions. The reader's attention is drawn to a large basic intrusive body that is shown (Fig. 4) in the region east of Great Bear Lake. This is the Muskox Complex, a boat-shaped layered intrusion studied by C. H. Smith (1962) and found by him to be similar to the Skaergaard and Stillwater complexes. The occurrence of this body in the midst of Mackenzie dyke intrusions, a similarity of K–Ar ages, and a concordance of remanent magnetism (Robertson, W. A., personal communication 1964) between the dykes and the complex suggest that these were related igneous events.

The Mackenzie dykes are rather wide, generally more than 100 ft. In coarse central parts of the dykes the plagioclase and pyroxene typically enclose a finer-grained pink matrix and very wide dykes typically contain reddish granophyric segregations. Biotite is megascopically visible in most of the wider members of the swarm. The chemical analysis (Table III) is the average of 20 analyses of samples from chilled margins of 20 Mackenzie dykes.

Burwash *et al.* (1963) reported the first K–Ar age determination made on these dykes. Their whole-rock K–Ar age (1 000–1 100 m.y.) is slightly less

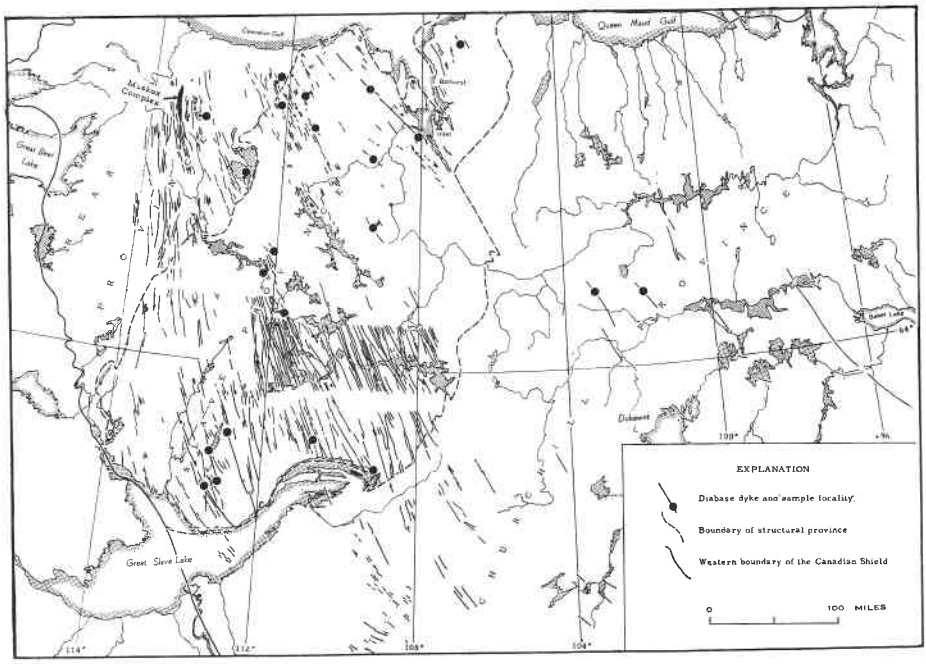


FIG. 4. Map showing the known distribution of Mackenzie dykes and the location of samples used in this study (apparent sudden changes in the density of dyke intrusion are due to variation in availability of aeromagnetic data).

than, but in fairly good agreement with, the K–Ar age of 1 315 m.y. reported by Fahrig and Wanless (1964). This age of 1 315 m.y. represents an average of five determinations from widely separated localities: four K–Ar biotite determinations and one whole-rock K–Ar determination. If the values of only the biotite determinations are averaged, the result is a K–Ar age of 1 295 m.y.

The Mackenzie dykes are expressed by strong positive anomalies on aeromagnetic maps, and so are easily traced in areas for which such maps are available.

Palaeomagnetic Data

This constitutes the first recorded palaeomagnetic data on dykes of the Mackenzie swarm. Forty-two oriented samples were collected from 28 widely separated localities. Eight of these samples were considered 'unstable' (see previous discussion of sample collection and measurement). The 23 collection sites of the remaining 34 samples are shown on Fig. 4 and the magnetic vectors after the samples were washed at 80 oersteds are shown on Fig. 5. The magnetic orientations of 3 of the 34 samples so treated appear to be anomalous.

SUDBURY DYKES

Geological Setting

The Sudbury dykes strike in a west-northwesterly direction across the Sudbury district. The swarm extends to the southeast where it appears to end

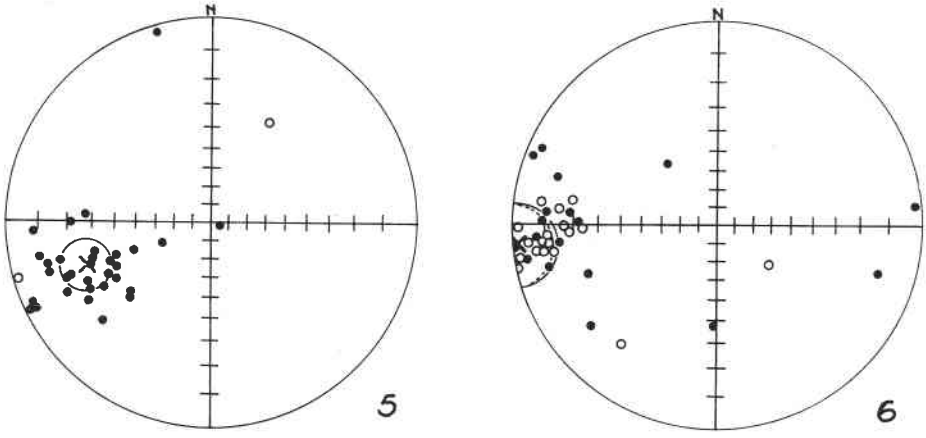


FIG. 5. Stereographic projection of magnetization directions, mean direction, and circle of confidence of Mackenzie dyke samples. ● North poles pointing downwards; ○, north poles pointing upward.

FIG. 6. Stereographic projection of magnetization directions, mean direction, and circle of confidence of Sudbury dykes. ● Downward pointing north poles; ○ upward pointing north poles.

at the boundary of the Grenville structural province; however, metamorphosed dykes, probably of this swarm, may be traced southeast into the Grenville. The extent of the swarm northwest of the Sudbury district is not known, largely because here the swarm is intermingled with another swarm of similar strike but of quite different age. The recognized dykes of this swarm are shown on Fig. 2.

The Sudbury dykes are generally more than 100 ft wide. They are characterized by a high content of olivine and by notable amounts of biotite. The chemical analysis above (Table III) is the average of 10 analyses of samples taken from the chilled margins of 10 Sudbury dykes.

Fahrig and Wanless (1964) have reported a K–Ar age of 1 220 m.y. for this swarm, but this is an average that includes dykes whose correlation is in doubt. Only two ages from biotites of undoubted Sudbury dykes are presently available. These K–Ar ages, 1 330 m.y. and 1 240 m.y., are concordant within the limits of analytical uncertainty and suggest a K–Ar age of about 1 285 m.y.*

Palaeomagnetic Data

P. J. Hood (1961), S. R. Sopher (1963), and D. W. Strangway (1961) have reported measurements of magnetization of the Sudbury dykes. Strangway collected from only one site. He reported the results of measurements of eight samples which were rather highly dispersed $k = 2.65$. Sopher (1963) reported measurements on seven samples collected at four sites. The dispersion of the magnetic vectors of his samples is quite low. In addition, to test Strangway's

*Fairbairn *et al.* have published a 1 020 m.y. K–Ar date on a Sudbury dyke (Fairbairn, H. W., Hurley, P. M., and Pinson, W. H., 1960. Mineral and rock ages at Sudbury – Blind River, Ontario. Proc. Geol. Assoc. of Canada, Vol. 12, pp. 41–66) and van Schmus a 1 205 ± 50 m.y. Rb/Sr date (van Schmus, W. R., 1964. Geochronology of the original Huronian area, Ontario, Canada. Abstract. Program of the 1964 annual meeting of the Geol. Soc. Amer., p. 211).

previously published hypothesis regarding the factor controlling the direction of magnetization in diabase dykes, Sopher sampled norite adjacent to a younger Sudbury dyke. He found that its magnetization was parallel to that of the dyke itself and concluded from this that the stable direction of magnetization in the dykes is parallel to the earth's field at the time of intrusion and is not controlled by some demagnetization process operating within the cooling dyke.

We have collected 45 samples from the sites shown on Fig. 2. Thirty-seven of these samples were stable and their magnetic vectors are shown on Fig. 6. The magnetization directions are reasonably consistent and their mean compares well with Sopher's mean for a much smaller number of measurements (Sopher 1963, Fig. 5, p. 13).

MOLSON DYKES

Geological Setting

The Molson dykes named from Molson Lake strike parallel to, and occur southeast of the boundary between, the Superior and Churchill structural provinces in northeastern Manitoba (Fig. 7).

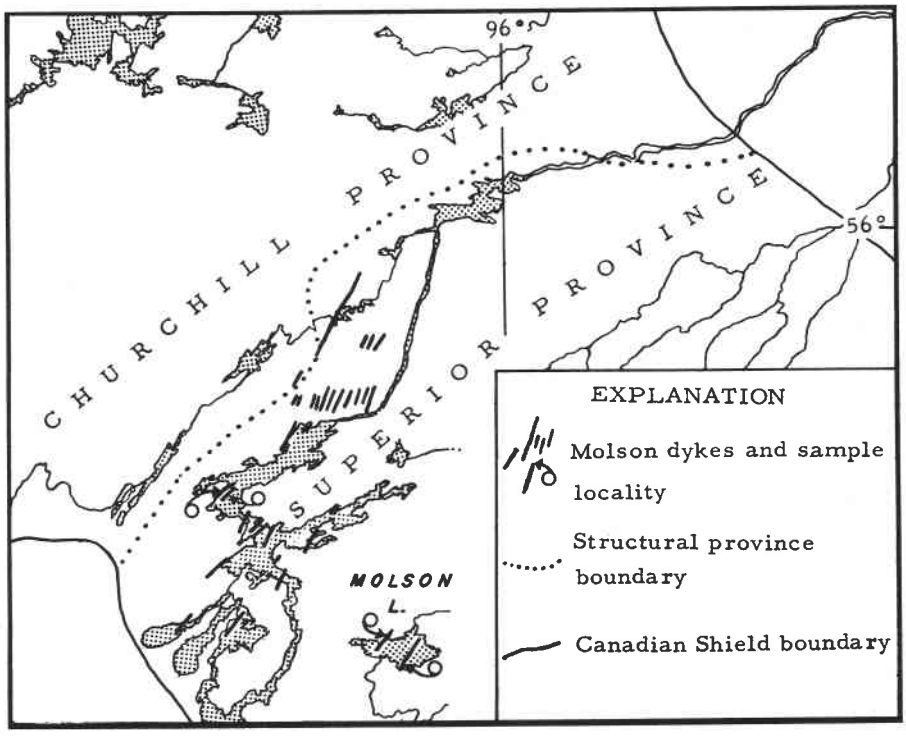


FIG. 7. Map showing the distribution of the Molson dykes and locations of palaeomagnetic samples.

Their known extent is limited to a belt about 100 miles wide in the Superior structural province. The length of individual dykes of the swarm is probably

much greater than shown in Fig. 8 as several have widths in excess of 200 ft and dykes of this width generally extend for several tens of miles.

The dykes of this swarm in the vicinity of the Superior–Churchill boundary have been slightly metamorphosed, perhaps suggesting an early Proterozoic age of intrusion. A single whole-rock K–Ar determination on a sample from Molson Lake yielded an age of 1 445 m.y. but the content of K in this rock was rather low (0.38%) for accurate determination. This K–Ar age is considered a first approximation to the age of intrusion.

These dykes are not expressed as pronounced anomalies on the aeromagnetic maps of the district.

Palaeomagnetic Data

These are the first recorded palaeomagnetic data on the dykes of this swarm. Seven samples were collected from five dykes. The directions of their natural remanent magnetization is relatively consistent, but magnetic washing in the laboratory increased the angle between magnetic vectors of cubes cut from the same sample. As a result, two samples were temporarily considered 'unstable'. The magnetization directions of the remaining five are plotted on Fig. 8, and

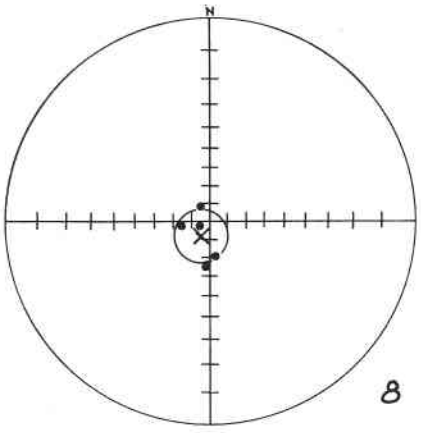


FIG. 8. Stereographic projection of magnetization directions, mean direction, and circle of confidence of Molson dyke samples. Solid circles are downward pointing north poles.

Fig. 7 shows the collection sites. Additional alternating field treatment with improved equipment on the seven samples and the study of additional samples may provide more consistent magnetization directions.

MARATHON DYKES

Geological Setting

The Marathon dykes are north of Lake Superior and trend in a northerly direction. They are named for the town of Marathon, one of the larger settlements of the area.

Their extent is difficult to assess because of the profusion of dykes of various

ages and trends in this area and because the Marathon dykes are generally narrow. The dykes at four of the five collection sites that yielded suitable samples had widths of less than 90 ft. Mapped dykes that probably belong to this swarm are shown on Fig. 2.

The chemical analysis reported in Table III is the average of four analyses of samples taken from chilled margins of four Marathon dykes.

A whole-rock K–Ar age of 1 810 m.y. has been obtained on a sample of dyke from this swarm. This should be considered a first approximation to the age of intrusion.

The dykes of this swarm are not expressed as pronounced anomalies on aeromagnetic maps of the region.

Palaeomagnetic Data

These are the first recorded palaeomagnetic data on Marathon dykes. Twelve samples of this swarm were collected from seven sites. One sample was found to be 'unstable', and two were anomalous. The five collection sites of the remaining nine samples are shown on Fig. 2 and the thermoremanent magnetization directions on Fig. 9.

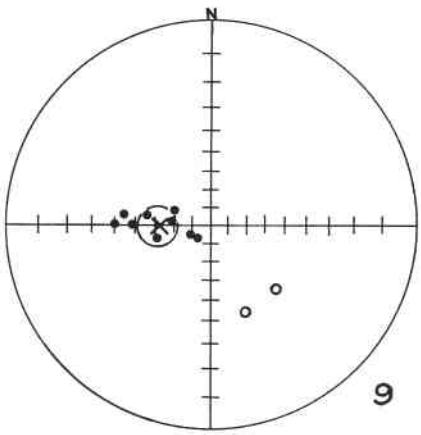


FIG. 9. Stereographic projection of magnetization directions, mean direction, and circle of confidence of Marathon dyke samples. ● North poles pointing downward; ○ north poles pointing upward.

The results of this preliminary study of the palaeomagnetics of the Marathon dykes is encouraging and further work is planned.

THE MATACHEWAN DYKES

Geological Setting

The Matachewan dykes form a northerly striking swarm extending from James Bay south to the 47th parallel, and from the Noranda area on the east to the Timmins area on the west. They are prominent in the gold camps of Ontario and the geological literature of these areas abounds in references to

'Matachewan type' dykes. Locally, for example in the Matachewan area of Ontario, these basalt dykes occur in greater profusion than do dykes of any other swarm anywhere else in the Shield. W. S. Dyer (1935) states, "the name Matachewan was first used by Miller for certain diabase dykes that cut the Algoman and earlier rocks but are overlain unconformably by the Cobalt conglomerate, because dykes of such relationships were first recognized in the Matachewan area by Burrows".

Most of the known, wider, Matachewan dykes are shown on Fig. 2. The dykes are generally less than 100 ft wide and frequently have a porphyritic or a glomeroporphyritic texture consisting of either individual crystals or roughly spherical aggregates of white or pale green-weathering plagioclase crystals in a finer-grained, generally dark green matrix. The analysis (Table III) is the average of eight analyses of samples of chilled marginal phases of eight Matachewan dykes. These dykes are referred to compositionally as quartz diabase.

The pre-Cobalt age of the Matachewan dykes provided an exception to the once held view that diabase dykes are intrusives of Keweenawan age and Gill and L'Esperance (1952) correctly used this as an argument to suggest that the Keewenawan age designation of most diabase was, in fact, not correct. Fahrig and Wanless (1963) reported a single K–Ar whole-rock age of 2 485 m.y. on the chilled margin of a Matachewan diabase dyke. A hornblende age of 1 740 $\pm$ 200 m.y. (early Proterozoic) has been reported by Campbell (1962) on a northerly striking dyke that cuts the Noranda mine. Our palaeomagnetic data suggest that this is a Matachewan dyke (Archaean). If this is correct, the age obtained by Campbell must be the result of post-intrusion argon loss from the dyke rock.

Matachewan dykes cause pronounced linear anomalies on the aeromagnetic maps of the region in which they occur.

Palaeomagnetic Data

Fifty-seven samples were collected from 25 sites on dykes thought to be Matachewan type. Four of these samples were considered unstable. The collection sites of the remaining 53 samples are shown on Fig. 2 and the magnetization directions after washing are plotted in Fig. 10 (*a* and *b*).

The Matachewan dykes appear to be divisible into two groups whose palaeomagnetic directions are roughly opposed. The magnetization direction of one group is to the northeast (Fig. 10*a*) and of the other group to the southwest (Fig. 10*b*). This relationship suggests the presence of a pole reversal during the formation of the dyke swarm.

Strangway published palaeomagnetic data on a few north–south dykes that confirm the mean magnetization directions reported here.

DISCUSSION OF PALAEOMAGNETIC DATA

1. In three of the six swarms described above (Mackenzie, Sudbury, Matachewan) the mean magnetization direction forms an angle of greater than

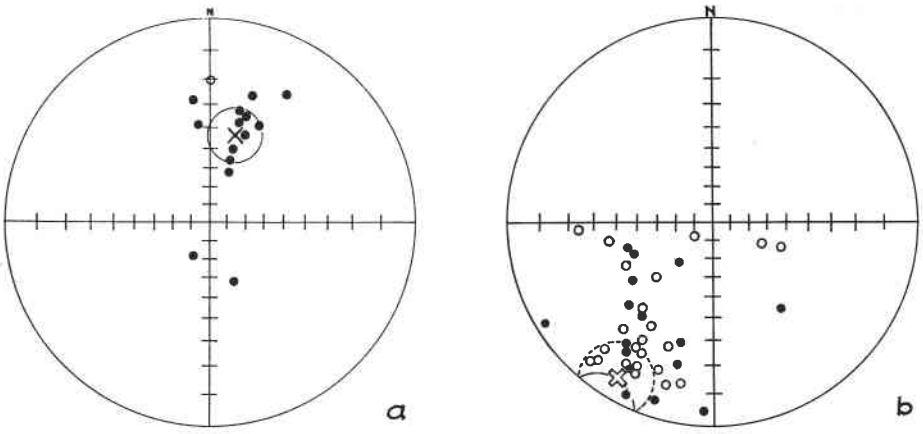


FIG. 10. (*a*) Stereographic projection of northeasterly pointing magnetization directions, mean direction, and circle of confidence of Matachewan dyke samples. ● Downward pointing north poles; ○ upward pointing north pole. (*b*) Stereographic projection of southwesterly pointing magnetization directions, mean direction, and circle of confidence of Matachewan dyke samples. ● North poles pointing downward; ○ north poles pointing upward.

30 degrees with the plane of the related dyke swarm. The angle between the dyke plane of the Mackenzie swarm and the mean vector is 65 degrees. These data contradict Strangway's (1961) suggestion that a correlation exists between palaeomagnetic and dyke directions such that the magnetization direction is within 24 degrees of the dyke plane. If the magnetization directions of dykes are parallel to the earth's field at the time of dyke crystallization, magnetization directions should be randomly oriented with respect to the planes of the dykes. If the magnetization directions were completely random, then it can be calculated that 40% of all mean vectors would lie within 24 degrees of any chosen plane. There is therefore a 40% chance that the mean vector of any dyke swarm will lie within 24 degrees of the plane of the dyke swarm. The correlation between vectors and dyke planes suggested by Strangway is not significant because of the small number of dyke swarms he studied.

2. The virtual pole positions (Table I) calculated from the mean vectors of the Mackenzie and Sudbury dykes are very close. This suggests that the two dyke swarms could have formed contemporaneously.

The magnetization directions of samples collected widely throughout some dyke swarms are moderately well grouped, which suggests that the intrusion of these swarms were igneous events of short duration. For example, dispersion of the magnetization directions within the Mackenzie swarm (Fig. 5) is moderately low and these samples were collected over an area of several hundred thousand square miles from dykes which vary in strike by at least 30 degrees. This suggests that not only was the intrusion of the Mackenzie dykes an event of short duration, but that a large section of the western Canadian Shield, including parts of three structural provinces, has been relatively stable except for epeirogenic movements for at least 1 300 million years.

3. In some samples the reason for the dispersion of magnetization directions within a swarm may be found in the geological setting of the swarm. For example, Fig. 11 shows the magnetization direction of 14 samples obtained

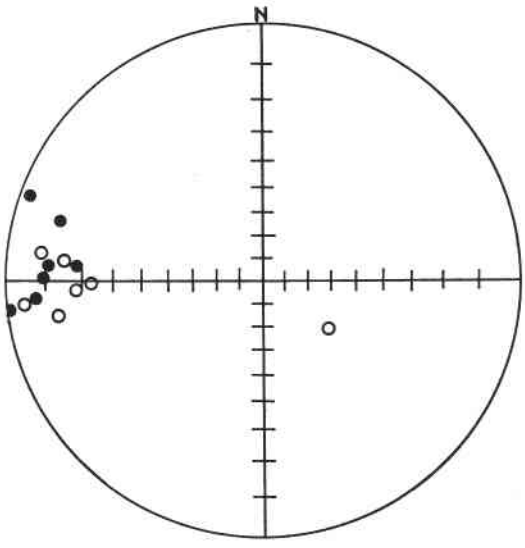


FIG. 11. Magnetization directions of 14 Sudbury dyke samples taken from 13 sites by a single collector. See text for explanation.

from 13 different sites in the Sudbury swarm by a single collector. The magnetization direction of one sample appears to be 'anomalous'. On closer study this sample was found to have certain chemical characteristics that distinguish it from the average Sudbury dyke. This rock contains 7% more SiO_2 and less than half the TiO_2 content of the average Sudbury dyke rock. Furthermore, the olivine and pyroxene typical of Sudbury dykes have been replaced in this rock by blue-green hornblende. In other respects, trend of the dyke, alkali content, and high biotite content, this is a typical Sudbury dyke. This sample was taken from within 10 miles of the contact between the Superior and Grenville structural provinces. This suggests that the alteration of pyroxene and olivine was due to the regional metamorphic effects of the Grenville orogeny rather than to deuteric processes. In certain cases, regional metamorphic effect might be assessed from a study of dispersion of the magnetization directions within a dyke swarm. This may prove to be a delicate thermometer as dyke samples exposed to different conditions of regional metamorphism may respond to slightly different alternating field treatment.

Differential rotation of dykes and dyke segments as a result of later crustal deformation may be an important factor in the dispersion of magnetization vectors within older dyke swarms. During an orogeny, a belt of craton adjacent to the mobile belt must experience severe stresses that result in innumerable movements along major and minor fault planes. The resultant jostling and

differential rotation of crustal blocks containing older dykes or segments of dykes may add a significant dispersion factor to the magnetization directions of the swarm. The Matachewan dykes, for example, are Archaean in age and predate both the Hudsonian and Grenville orogenies. The south part of the Matachewan swarm lies within 100 miles of rocks that were intensely deformed during both orogenies. It is likely that faulting related to these periods of mountain building contributed to the dispersion of the magnetization directions of the Matachewan dykes.

We intend to evaluate the dispersion of magnetization vectors in different parts of dyke swarms as a possible index of post-dyke regional deformation.

CHEMISTRY OF THE DYKES

The chemical and mineralogical as well as the palaeomagnetic characters of diabase dykes may be useful guides in solving problems of correlation of diabase dykes and of dyke swarms. Chemical and mineralogical work on the samples is being carried out but as this will not be completed for some time, a summary of chemical data that is particularly pertinent to the present palaeomagnetic study is presented here. The quality of the analyses included in averages 1–5, Table III, is not discussed here; however, the writers feel that their comparability with the results of other analysis and their duplicability are well within the range necessary in making the generalizations below.

It is widely accepted that the diabases of continental dyke and sill swarms are strictly the intrusive equivalents of tholeiitic flood basalts (Turner and Verhoogen 1960, p. 209). Our preliminary chemical work has indicated that this is not so. The alkali–silica ratios (Fig. 12) of the Sudbury and possibly of the Mackenzie swarms place them in the alkalic basalt field as defined by

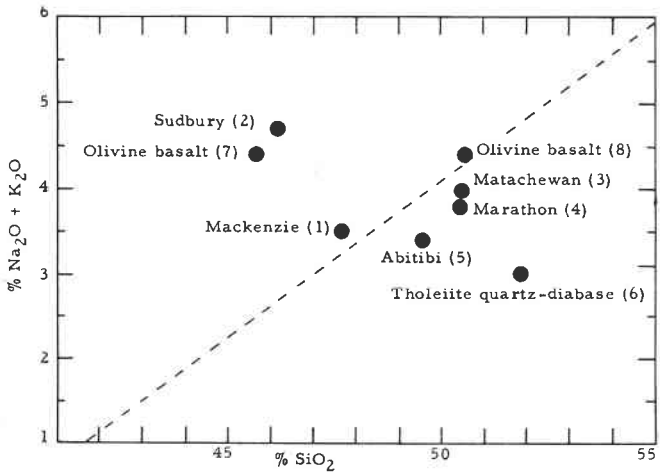


FIG. 12. Alkali–silica diagram of Canadian Shield diabase dyke swarms and comparable rock types from Table II. The diagonal broken line is from MacDonald and Katsura (1964) and in their study separates Hawaiian rocks with tholeiitic mineral composition (lower right field) from rocks with alkalic mineral compositions (upper left field).

MacDonald and Katsura (1964). In addition, titanium is notably higher in the Mackenzie and Sudbury dykes, a feature of practical value for distinguishing between alkalic and tholeiitic basalts. The titanium-rich nature of the Sudbury dykes is indicated by the presence of strongly purplish titanaugite. The Sudbury dykes are certainly typical alkalic basalts.

This has an important bearing on one facet of our palaeomagnetic investigation, namely, the possible correlation of the Sudbury and Mackenzie dykes. The two swarms are not significantly different in terms of both palaeomagnetic pole position and K–Ar age (Table I). Furthermore, the Mackenzie dykes have been traced southeast from the area shown in Fig. 4, across the western Canadian Shield to within a few hundred miles of Sudbury dykes of similar trend. The Sudbury dykes are decidedly more undersaturated than the Mackenzie dykes; however, our preliminary chemical data suggest that they belong to the same great alkalic suite of rocks. The two swarms will be examined for evidence of chemical convergence in passing along strike from one swarm to the other.

CONCLUSIONS

(1) The close grouping of remanent magnetization directions of samples collected widely throughout certain Precambrian basaltic dyke swarms suggests that the dykes of individual swarms were intruded over a geologically short period of time and acquired and retained a stable thermoremanent magnetization. The general absence of evidence of pole reversal suggests relatively rapid formation of the dyke swarms.

(2) The lack of evidence of a relationship between the direction of magnetization and the orientation of the plane of the dyke supports the usual assumption that the magnetization was parallel to the earth's magnetic field at the time of the cooling of the dyke.

(3) These facts make palaeomagnetic data of practical value in correlating diabase dykes and in correlating diabase dykes with intrusive and extrusive equivalents. For example, the pole positions of two swarms, the Mackenzie and Sudbury, are identical within limits of analytical uncertainty.

(4) Preliminary pole positions have been determined for six Precambrian diabase dyke swarms (Table I and Fig. 13).

(5) Some data on palaeomagnetic dyke data may be useful in distinguishing between late magmatic (deuteric) effects within dykes and the effects of low grade regional metamorphism on dykes. Caution should be used in sampling dykes near the boundaries of younger structural provinces.

(6) The dispersion of magnetization directions within a swarm of basalt dykes is greater than that generally recorded in single basic volcanic layers or in single basic intrusive bodies. This is attributed to the great chance of lightning striking high dyke ridges; to the greater period of time involved in the intrusion of a dyke swarm; to the difficulty often encountered in correctly assigning dykes to a single swarm; to low-grade regional metamorphic effects; and to differential rotation of dykes and dyke segments during later crustal deformation.

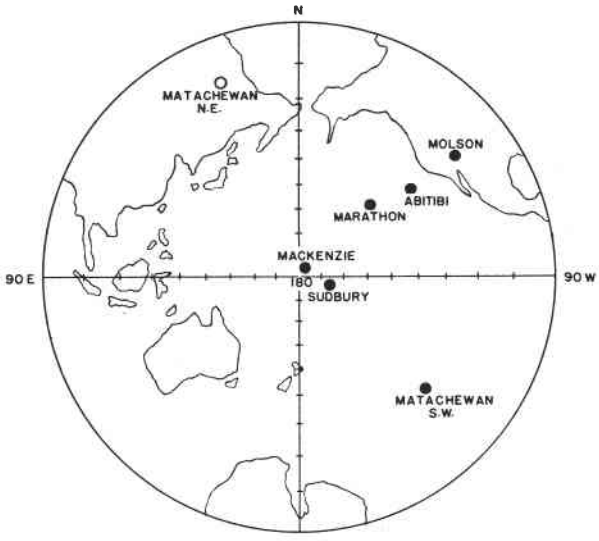


FIG. 13. Preliminary pole positions of six Precambrian dyke swarms.

(7) The more extensive diabase dyke swarms are probably indicative of great tensional zones and may be considered per se to be evidence of the small-scale disruption of continents and evidence of continental drift.

(8) A requirement of any continental drift hypothesis is the correlation of ancient diabase dyke swarms between segments of the hypothetical disrupted continent. The palaeomagnetic poles of dyke swarms from different continental blocks should be determined to see whether palaeomagnetic poles coincide when continental blocks are rotated back into their suggested primordial positions.

(9) Not all diabase dyke swarms are chemical intrusive equivalents of tholeiitic flood basalts. This is significant in terms of the origin of basalt and greatly enhances the possibility of correlating dyke swarms and correlating dykes with extrusive equivalents.

REFERENCES

BURWASH, R. A., BAADSGAARD, H., CAMPBELL, F. A., CUMMING, G. L., and FOLINSBEE, R. E. 1963. Potassium–argon dates of diabase dyke systems, District of Mackenzie, N.W.T. The Canadian Mining and Metallurgical Bulletin, Transactions, Vol. LXVI. pp. 303–307.

CAMPBELL, F. A. 1962. Age of mineralization at Quemont and Horne mines. Can. Min. Metallurgical Bull. **55**, No. 605, 627.

DYER, W. S. 1935. Ont. Dept. Mines Annual Rept. Vol. XLIV, pt. II.

FAHRIG, W. F. and WANLESS, R. K. 1963. Age and significance of diabase dyke swarms of the Canadian Shield. Nature, **200**, No. 4910, 934.

FISHER, R. A. 1953. Dispersion on a sphere. Phil. Trans. Roy. Soc. London, Ser. A, **217**, 295.

GILL, J. E. and L'ESPERANCE, R. 1952. Diabase dykes in the Canadian Shield. Trans. Roy. Soc. Can. Sect. IV, Ser. 3, **46**, 25.

HOOD, P. J. 1961. Palaeomagnetic study of the Sudbury Basin. J. Geophys. Res. **66**, No. 4, 1235.

LAROCHELLE, A. 1962. Palaeomagnetism of the Monteregian Hills, S.E. Quebec. Can. Dept. Mines Tech. Surv. Geol. Surv. Can., Geol. Surv. Bull. No. 90.

MACDONALD, G. A. and KATSURA, T. 1964. Chemical composition of Hawaiian lavas. J. Petrol. **5**, pt. 1, 82.

SMITH, C. H. 1962. Notes on the Muskox intrusion, Coppermine River area, District of Mackenzie. Can. Dept. Mines Tech. Surv. Geol. Surv. Can. Paper 61-25.

SOPHER, S. R. 1963. Palaeomagnetic study of the Sudbury irruptive. Can. Dept. Mines Tech. Surv. Geol. Surv. Can., Geol. Surv. Bull. No. 90.

STOCKWELL, C. H. 1962. A tectonic map of the Canadian Shield. *In* The tectonic of the Canadian Shield. *Edited by* J. S. Stevenson. Roy. Soc. Can., Spec. Publ. No. 4.

STRANGWAY, D. W. 1960. Magnetic properties of some Canadian diabase dyke swarms. Ph.D. Thesis, University of Toronto.

STRANGWAY, D. W. 1961. Magnetic properties of diabase dykes. J. Geophys. Res. **66**, No. 9, 3021.

TURNER, F. J. and VERHOOGEN, J. 1960. Igneous and metamorphic petrology. 2nd ed. (McGraw-Hill, New York).

WATSON, G. S. and IRVING, E. 1956. Statistical methods in rock magnetism. Monthly Notices Roy. Astron. Soc., Geophys. Suppl. No. 7, 289.